



Quantum optimal control in quantum technologies. Strategic report on current status, visions and goals for research in Europe

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Abstract

Quantum optimal control (QOC) is a powerful tool for the design of quantum control pulses that accomplish a given task. In this report, we provide a comprehensive overview of the field of QOC, its current status, and its future prospects. We discuss the various applications of QOC in quantum technology, from quantum computing to quantum communication, and we highlight the challenges and opportunities in this field. We also provide a list of open problems and a roadmap for future research.

Keywords: Quantum optimal control, Quantum control, Quantum computing, Quantum communication, Quantum technology

1. Introduction

Quantum optimal control (QOC) refers to a set of methods to design and optimize quantum control pulses that accomplish a given task. The design of quantum control pulses is a central problem in quantum technology, and it has been the subject of extensive research in the last few decades [1–4]. In this report, we provide a comprehensive overview of the field of QOC, its current status, and its future prospects. We discuss the various applications of QOC in quantum technology, from quantum computing to quantum communication, and we highlight the challenges and opportunities in this field. We also provide a list of open problems and a roadmap for future research.

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